3.7 Graph of Variations

Variables:



Every day, Harini travels from her home, cycling at a uniform speed, to reach her school. You can state this mathematically by an equation $d = rt$, where d stands for distance travelled at any time t and r is the uniform rate of speed.

Suppose you want to find the distance covered by her at a speed of 20 *km* per hour when she has cycled for fifteen minutes.

$r = 20$ and $t = \frac{1}{4}$ (how?) and we find d to be $rt = 20 \times \frac{1}{4} = 5$ km.

Here, we say that d is a dependent variable and r and t are independent variables. As the distance d travelled depends upon the rate r and time used t.

Thus, an independent variable represents a quantity that is manipulated in a given situation where as a dependent variable represents a quantity whose value depends on how the independent variable is manipulated.



Equations that describe the relationship between two variables in a sentence express the variation between those variables.

Consider the monthly income of Server Suresh who works in a hotel where he is paid ₹ 50 per hour.

There are two variables here. One is the monthly income and the other is the number of hours he works. Which among the two is the independent variable?

Constants:

You know how to calculate the area of a circle when the length of its radius is given. If the area required is A and the length of radius is r, then the formula

$$A = \pi r^2$$

gives the required result. Here, the area A depends upon the length r of radius; thus A is a dependent variable and r is the independent variable. But what can we say about π? It is a number that remains the same in all the situations. It is constant.

A constant is a quantity that assumes a fixed value throughout in a specific mathematical context.

Two types of variation:

When two things are in proportion, there is a relation between them, due to which, if the value of one of them changes, the value of the other also changes. We look into two types of variations here:

(i) Direct variation

(ii) Indirect variation.

(i) Direct variation:

When you go to the market, to buy more apples, you'll have to spend more amount of money. If the cost of one kg of apples is ₹ 200, you pay as follows:

Weight (Kg)	1	2	3	4	5
Cost (₹)	200	400	600	800	1000

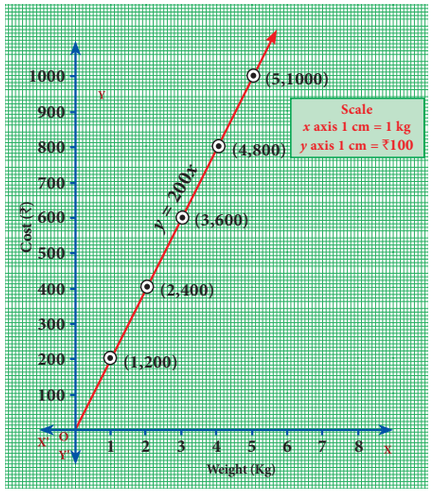


Fig. 3.8

3 Algebra.indd 123 16-12-2020 01:06:23 PM